$$
\begin{aligned}
T_{L\leftarrow R} &= T_L^{-1} T_R \\
\widetilde{T}_L &= S T_L, \qquad \widetilde{T}_R = S T_R \\
\widetilde{T}_L^{-1} \widetilde{T}_R &= (S T_L)^{-1} (S T_R) \\
&= T_L^{-1} S^{-1} S T_R \\
&= T_L^{-1} T_R
\end{aligned}
$$